In this pattern we can replace X by "ophthalmologist" and Z by "eye doctor" to obtain the original argument. But here is another argument of the same form:

> Mei is either a mathematician or a botanist.
> Mei isn't a botanist.
> ∴ Mei is an acrobat.

This argument is clearly not valid, since we can imagine a mathematician named Mei who is not an acrobat.

Our strategy as logicians will be to come up with a notion of "case" on which an argument turns out to be valid if it is formally valid. Clearly such a notion of "case" will have to violate not just some laws of nature but some laws of English. Since the first argument is invalid in this sense, we must allow as counterexample a case where Priya is an ophthalmologist but not an eye doctor. That case is not a conceivable situation: it is ruled out by the meanings of "ophthalmologist" and "eye doctor."

When we consider cases of various kinds in order to evaluate the validity of an argument, we will make a few assumptions. The first assumption is that every case makes every sentence true or not true—at least, every sentence in the argument under consideration. That means first of all that any imagined scenario which leaves it undetermined if a sentence in our argument is true will not be considered as a potential counterexample. For instance, a scenario where Priya is a dentist but not an ophthalmologist will count as a case to be considered in the first few arguments in this section, but not as a case to be considered in the last two: it doesn't tell us if Mei is a mathematician, a botanist, or an acrobat. If a case doesn't make a sentence true, we say it makes it FALSE. We'll thus assume that cases make sentences true or false but never both.[1]

[1]Even if these assumptions seem common-sensical to you, they are controversial among philosophers of logic. First of all, there are logicians who want to consider cases where sentences are neither true nor false, but have some kind of intermediate level of truth. More controversially, some philosophers think

2.4 Sound arguments

Before we go on and execute this strategy, a few clarifications. Arguments in our sense, as conclusions which (supposedly) follow from premises, are of course used all the time in everyday and scientific discourse. When they are, arguments are given to support or even prove their conclusions. Now, if an argument is valid, it will support its conclusion, but *only if* its premises are all true. Validity rules out the possibility that the premises are true and the conclusion is not true at the same time. It does not, by itself, rule out the possibility that the conclusion is not true, period. In other words, it is perfectly possibly for a valid argument to have a conclusion that isn't true!

Consider this example:

Oranges are either fruit or musical instruments.
Oranges are not fruit.
∴ Oranges are musical instruments.

The conclusion of this argument is ridiculous. Nevertheless, it follows from the premises. *If* both premises are true, *then* the conclusion just has to be true. So the argument is valid.

Conversely, having true premises and a true conclusion is not enough to make an argument valid. Consider this example:

London is in England.
Beijing is in China.
∴ Paris is in France.

The premises and conclusion of this argument are, as a matter of fact, all true, but the argument is invalid. If Paris were to declare independence from the rest of France, then the conclusion would no longer be true, even though both of the premises would remain true. Thus, there is a case where the premises of this argument

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we should allow for the possibility of sentences to be both true and false at the same time. There are systems of logic in which sentences can be neither true nor false, or both, but we will not discuss them in this book.